Aman K Gupta

AIML Engineer | Generative AI Engineer | LLM Engineer
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Summary

Final-year B.Tech student (AI & ML, graduating 2027) who builds LLM-powered features end to end: Python backend, REST APIs, a web frontend, SQL databases and AI parts such as prompts, agents, RAG and model APIs. I measure the quality of AI output with evaluations, write automated tests, and ship through Git and CI.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, OpenCV, YOLO, MediaPipe, RNN/GRU, MLflow
Generative AI & LLMs: LLMs, LLM APIs, Prompt Engineering, RAG, Hugging Face Transformers, LoRA/QLoRA, PEFT, Unsloth, vLLM, Quantization, LangChain, LangGraph, Embeddings, Vector Databases, Ragas, Agentic AI
AI Evaluation: LLM Output Evaluation, Test Sets, Scoring & Comparing Versions, Ragas, Retrieval Metrics (hit@1)
Backend, APIs & Web: FastAPI, REST APIs, HTTP, WebSockets, React, Next.js, Asynchronous Processing, Distributed Systems
Databases & Messaging: PostgreSQL, MongoDB, Redis, Qdrant, RabbitMQ, Celery, Relational Database Design
Cloud, DevOps & Testing: AWS, GCP, Docker, Kubernetes, Git, Pull Requests, Code Review, Automated Testing, GitHub Actions (CI/CD), Prometheus, Grafana, Linux
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Experience

Introspect Labs Jun 2026 – Jul 2026
Generative AI Engineer Intern *Remote*
- Built **PersonaMatrix**, a real-time AI avatar service that turns text/voice into emotion-aware, lip-synced video, using modular STT–LLM–TTS providers with Python, FastAPI, Socket.IO, PyTorch and MediaPipe.
- Cut per-turn latency from ∼80s to 17s (4.5x) using transport profiling, SHA-keyed caching, Opus audio and streamed LLM output; added provider fallbacks to keep the service running when a model API fails.
- Wrote 55 automated tests and set up CI, so each change was tested before it shipped.
- Checked AI output with numbers: trained a context-conditioned GRU-CVAE on identity-gated footage (91% context separability) and built a FACS/blendshape harness that measured 94% emotion recognition vs. a 17% chance baseline.
- **Stack:** Python, FastAPI, Socket.IO, PyTorch, MediaPipe, GitHub Actions.

DOI Soft Tech Jun 2025 – Aug 2025
AI Engineer Intern (AI Developer | Chatbot Developer) *Remote*
- Developed an LLM-powered Python/JavaScript browser extension that automated social-media content generation, saving 10+ team hours/week.
- Built and deployed supervised/unsupervised ML models, reducing data-pipeline latency by ∼40% through feature engineering and inference optimization.
- Deployed containerized AI services on AWS/GCP with Docker/Kubernetes; integrated REST/n8n workflows and analyzed 50K+ records using SQL, Pandas and NumPy.
- **Stack:** Python, JavaScript, SQL, Pandas, NumPy, Docker, Kubernetes, AWS, GCP, n8n.

IBM Jun 2026 – Jul 2026
Machine Learning Engineer *Remote*
- Developed and optimized deepfake detection models using PyTorch, TensorFlow, OpenCV and Python, improving reported detection accuracy from 83% to 95%.
- Implemented and compared deep learning architectures from 3–4 research papers, evaluating model performance using accuracy, precision and recall.
- Integrated MediaPipe, ArcFace and facial feature extraction techniques to support computer vision model development and experimentation.
- **Stack:** Python, PyTorch, TensorFlow, OpenCV, MediaPipe, ArcFace.

Projects

RAGForge | *Multi-tenant RAG-as-a-Service Platform* | GitHub Link | Demo Video
- **Problem:** teams need a simple way to add answers with sources from their own documents. Built a full-stack, multi-tenant RAG-as-a-Service platform with a Python backend, streaming chat API, page-level citations, an embeddable JS widget and a dashboard.
- **How I checked quality:** measured retrieval with a test set; hybrid search + reranker raised hit@1 from 87% to 97%.
- Built a fault-tolerant ingestion pipeline (transactional outbox, RabbitMQ retries, dead-letter queue) and a two-level Redis + semantic cache that cut repeat-answer latency from 1.3 s to 18 ms.
- Found and fixed two bottlenecks with k6 load tests: a DB connection-pool deadlock (errors 92% → 0% at 50 users) and reranker memory crashes (0 errors at 20 users; p50 2.0 s → 1.3 s).
- Wrote ∼390 automated tests that run in GitHub Actions CI; deployed on AWS EC2 with Docker Compose and Caddy (HTTPS); added Prometheus/Grafana metrics and OpenTelemetry tracing.
- **Stack:** Python, FastAPI, PostgreSQL, Redis, RabbitMQ, JavaScript, Docker, AWS, GitHub Actions.

RishiAI | *Domain-Specific LLM Fine-Tuning & RAG System* | GitHub
- **Problem:** general LLMs answer Vedic and Ayurvedic questions poorly. Built 8 automated web scrapers and a PDF pipeline to curate 7,128 documents (10.7M tokens) with verse-level metadata for training and retrieval.
- **Model:** fine-tuned Qwen2.5-14B-Instruct using 4-bit QLoRA, rank-64 rsLoRA and Unsloth on Modal L40S GPUs (275M trainable parameters, 4,096-token sequences).
- Deployed the model with vLLM on Modal (16K-token context window); published weights to Hugging Face and cut model storage by 70% (29.5 GB → 9 GB) with GGUF quantization.
- **How I checked quality:** built a two-stage RAG pipeline (37.7K chunks, BGE embeddings, MongoDB Atlas Vector Search, cross-encoder reranking) and scored it with Ragas for faithfulness, relevance and retrieval quality.
- **Stack:** Python, PyTorch, Qwen2.5-14B, QLoRA, Unsloth, vLLM, Modal, MongoDB Atlas, Hugging Face, Ragas.

Education

University of Petroleum and Energy Studies (UPES) 2023 – 2027
B.Tech in Computer Science Engineering – Specialization in AI & ML *Dehradun, India*